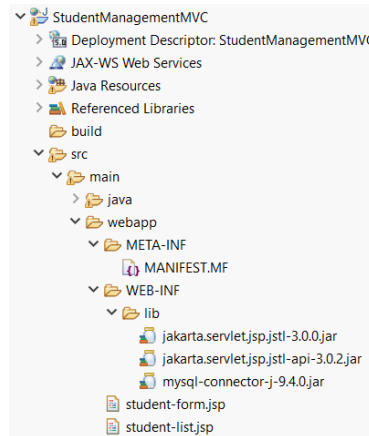


WEB APPLICATION DEVELOPMENT

LAB 5 – INCLASS REPORT



Project Structure

I. View the student list flow (example):

1. Browser → GET /student?action=list
2. StudentController receives request
3. Controller calls StudentDAO.getAllStudents()
4. DAO queries database, returns List<Student>
5. Controller sets attribute: request.setAttribute("students", list)
6. Controller forwards to student-list.jsp
7. JSP displays students from attribute
8. HTML response sent to browser

Student Management System (MVC)					
+ Add New Student					
ID	Student Code	Full Name	Email	Major	Actions
26	SV001	John Smith	john.smith@email.com	Computer Science	Edit Delete
22	SV016	John Doe	john@email.com	Computer Science	Edit Delete
21	SV015	Lucas Tran	lucas.tran@email.com	Computer Science	Edit Delete
20	SV014	Charlotte Do	charlotte.do@email.com	Software Engineering	Edit Delete
19	SV013	Benjamin Le	benjamin.le@email.com	Data Science	Edit Delete
18	SV012	Mia Vo	mia.vo@email.com	Information Technology	Edit Delete
17	SV011	William Hoang	william.h@email.com	Computer Science	Edit Delete
16	SV010	Ava Pham	ava.pham@email.com	Software Engineering	Edit Delete
15	SV009	Daniel Kim	daniel.k@email.com	Data Science	Edit Delete
14	SV008	Sophia Tran	sophia.tran@email.com	Computer Science	Edit Delete
13	SV007	James Lee	james.lee@email.com	Software Engineering	Edit Delete
12	SV006	Olivia Nguyen	olivia.n@email.com	Information Technology	Edit Delete
5	SV005	David Wilson	david.w@email.com	Computer Science	Edit Delete
4	SV004	Sarah Davis	sarah.d@email.com	Data Science	Edit Delete
3	SV003	Michael Brown	michael.b@email.com	Software Engineering	Edit Delete

Display student list

II. Add new student flow (Create):

1. Browser → GET /student?action=showForm: When the users click “Add New Student” button on student-list.jsp, the browser sends a GET request with action=showForm.
2. StudentController receives the request: The StudentController servlet’s doGet() method is executed, check the action parameter, call showForm() method.

3. Controller forwards to student-form.jsp: The showForm() method simply forwards the request to the form page via request.getRequestDispatcher("student-form.jsp"); and no database access occurs here.
4. JSP form displayed in browser
5. User fills out form and submits
6. StudentController receives POST /student?action=insert
7. Controller extracts form data
8. Controller calls DAO to insert student
9. DAO inserts into database
10. Controller redirects to list view
11. Browser → GET /student?action=list
12. StudentController → DAO.getAllStudents()
13. student-list.jsp displays updated list

Add New Student

Student Code:

SV017

Full Name:

John Daniel

Email:

john.daniel@email.com

Major:

Information Technology

 Save

Cancel

Student Management System (MVC)

Student added successfully

 Add New Student

ID	Student Code	Full Name	Email	Major	Actions
27	SV017	John Daniel	john.daniel@email.com	Information Technology	 Edit  Delete

Adding student successfully

III. Update student information:

1. Browser → GET /student?action=edit&id={studentId}
2. StudentController receives request
3. Controller retrieves student by ID
4. Controller forwards to student-form.jsp
5. JSP displays pre-filled form
6. User edits and submits the form
7. StudentController receives POST /student?action=update
8. Controller extracts form data
9. Controller calls DAO to update record
10. Controller redirects back to list
11. Browser → GET /student?action=list
12. Controller forwards to student-list.jsp

Edit Student

Student Code:

SV017

Full Name:

John Daniel

Email:

john.daniel@gmail.com

Major:

Information Technology

 Update

Cancel

Student Management System (MVC)

Student updated successfully

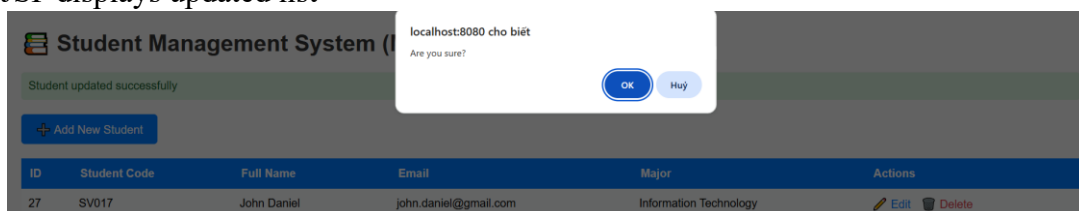
 Add New Student

ID	Student Code	Full Name	Email	Major	Actions
27	SV017	John Daniel	john.daniel@gmail.com	Information Technology	 Edit  Delete

Updating student successfully

IV. Delete student:

1. Browser → GET /student?action=delete&id={studentId}
2. StudentController receives the request
3. Controller extracts student ID
4. Controller calls DAO.deleteStudent(id)
5. DAO completes, returns to controller
6. Controller redirects to list view
7. Browser → GET /student?action=list
8. Controller calls DAO.getAllStudents()
9. Controller forwards to student-list.jsp
10. JSP displays updated list



Student Management System (MVC)

Student deleted successfully

 Add New Student

ID	Student Code	Full Name	Email	Major	Actions
26	SV001	John Smith	john.smith@email.com	Computer Science	 Edit  Delete

Deleting student successfully